ENGR 3323: Signals and Systems

Ch6
Continuous-Time Signal Analysis

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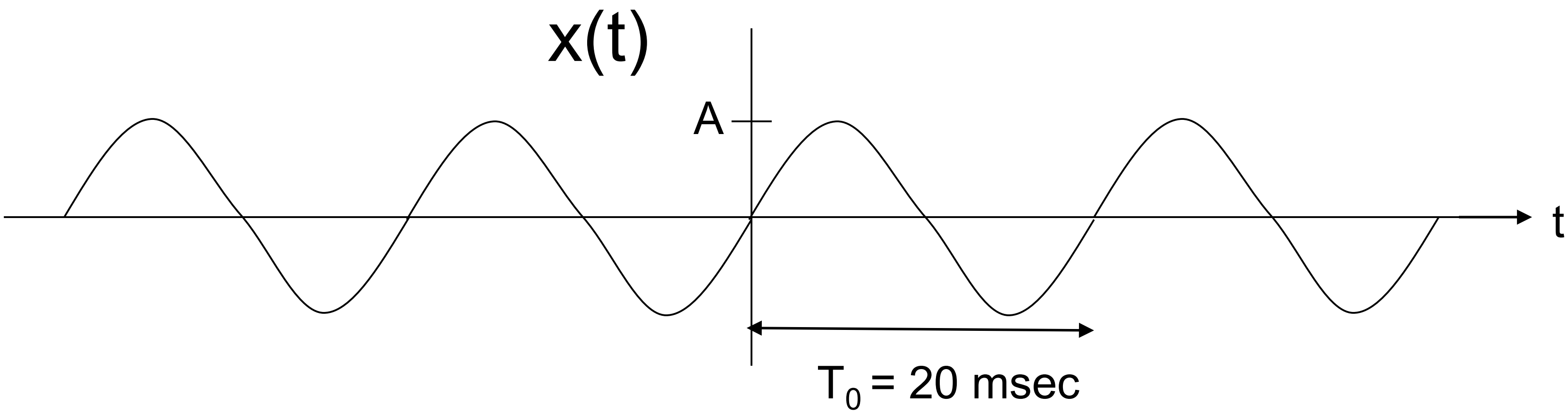
Outline

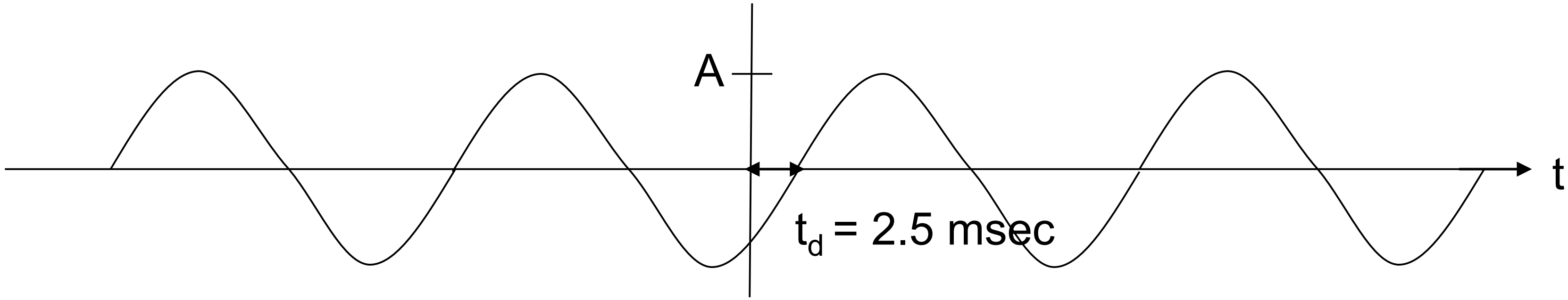
- Introduction
- Fourier Series (FS) Representation of Periodic Signals.
- Trigonometric and Exponential Form of FS.
- Gibbs Phenomenon.
- Parseval’s Theorem.
- Simplifications Through Signal Symmetry.
- LTIC System Response to Periodic Inputs.

Sinusoidal Wave and phase

$x(t) = A\sin(\omega t) = A\sin(2\pi 50t)$



$x(t-0.0025)= A\sin(2\pi 50[t-0.0025])$
$= A\sin(2\pi 50t-0.25\pi)= A\sin(2\pi 50t-45^o)$



Time delay t_d = 2.5 msec correspond to phase shift $\theta = 45^o$

Representation of Quantity using Basis

- Any number can be represented as a linear sum of the basis number {1, 10, 100, 1000}

Ex: 10437 =10(1000) + 4(100) + 3(10) +7(1)

- Any 3-D vector can be represented as a linear sum of the basis vectors {[1 0 0], [0 1 0], [0 0 1]}

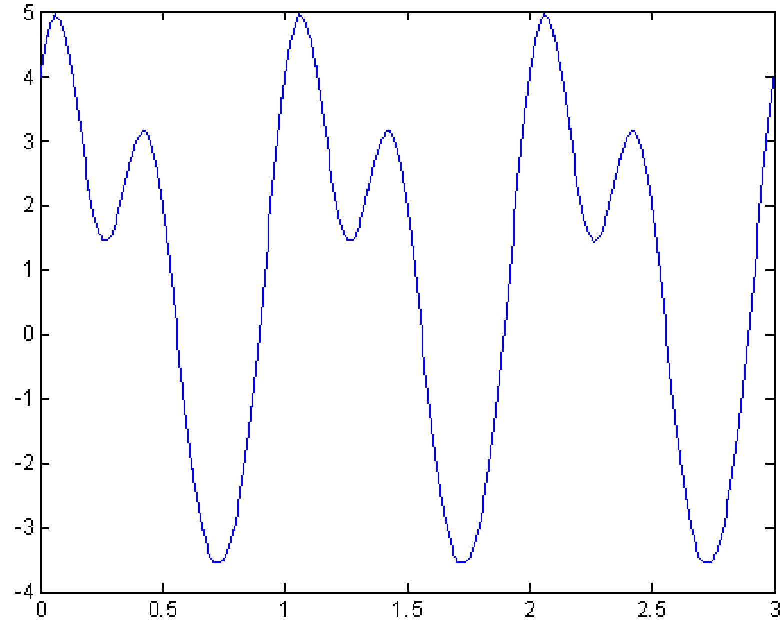
Ex: [2 4 5]= 2 [1 0 0] + 4[0 1 0]+ 5[0 0 1]

Basis Functions for Time Signal

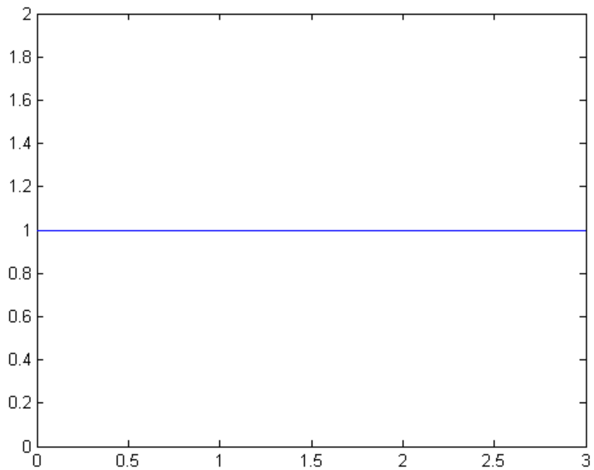
- Any periodic signal $x(t)$ with fundamental frequency ω_0 can be represented by a linear sum of the basis functions **$\{1, \cos(\omega_0 t), \cos(2\omega_0 t), \ldots, \cos(n\omega_0 t), \sin(\omega_0 t), \sin(2\omega_0 t), \ldots, \sin(n\omega_0 t)\}$**

Ex:

$$x(t) = 1 + \cos(2\pi t) + 2\cos(2\pi 2t) + 0.5\sin(2\pi 3t) + 3\sin(2\pi t)$$



$$x(t) = 1 + \cos(2\pi t) + 2\cos(2\pi 2t) + 3\sin(2\pi t) + 0.5\sin(2\pi 3t)$$



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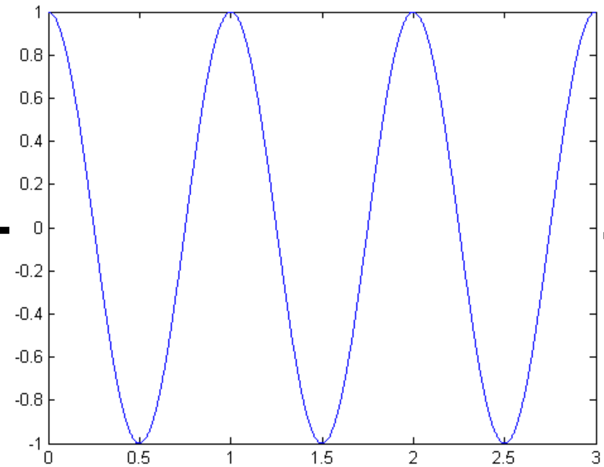

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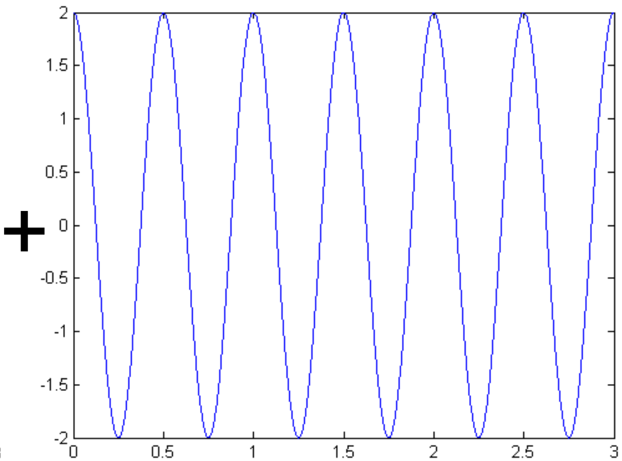

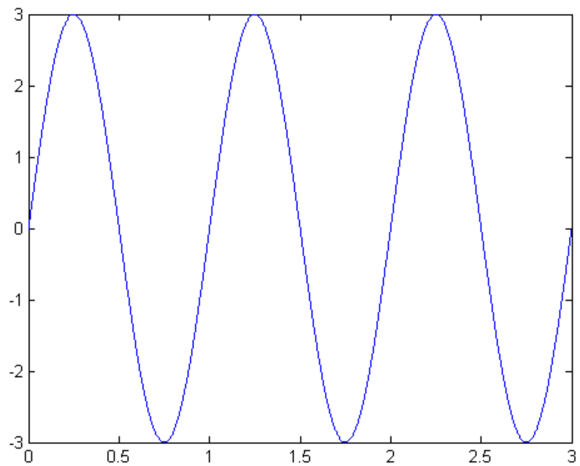

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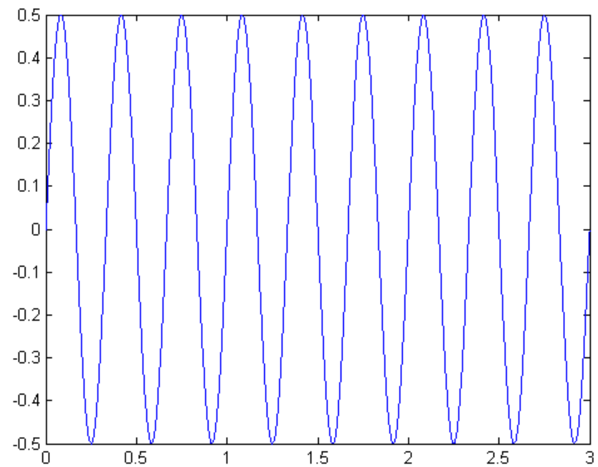

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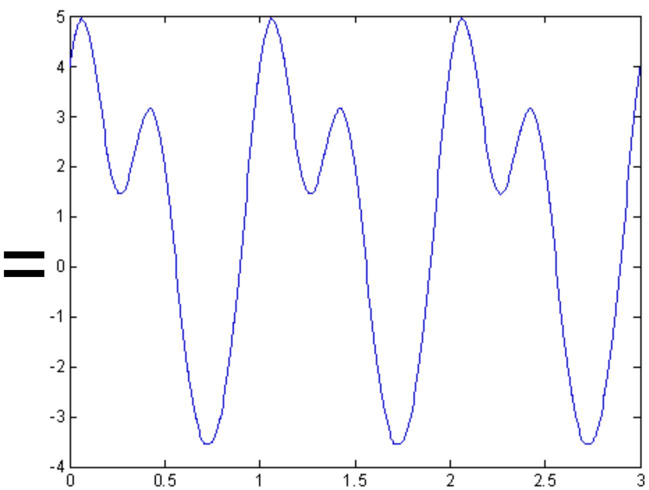
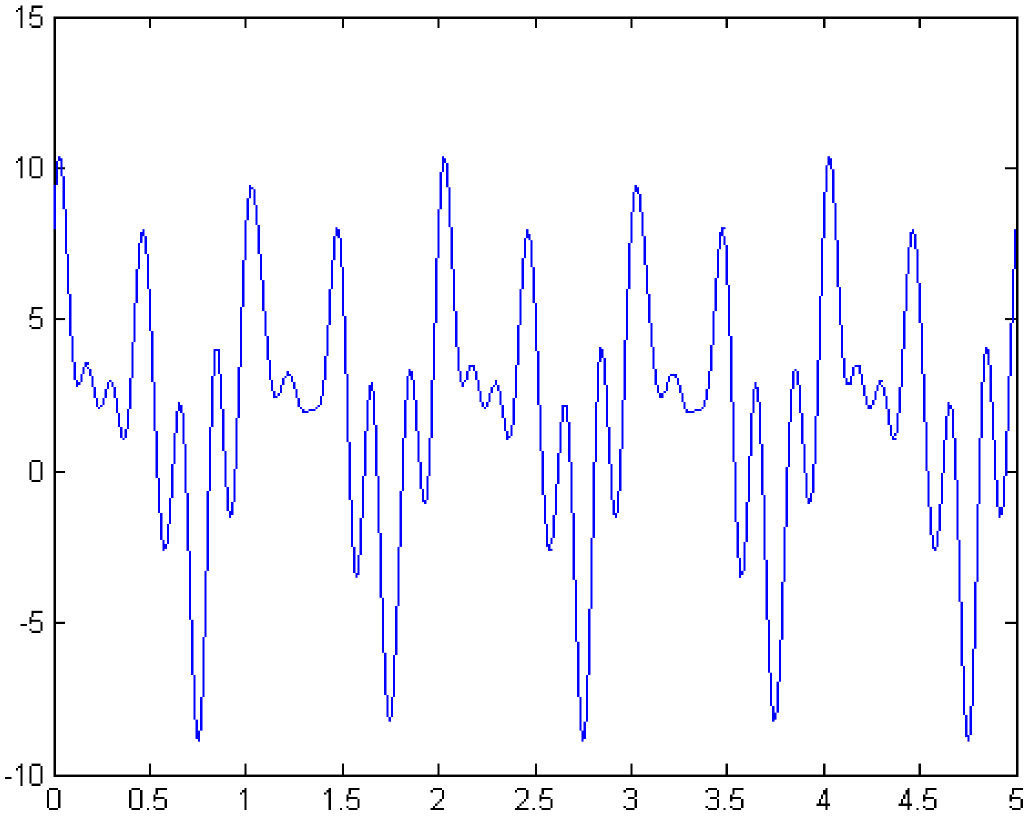


Purpose of the Fourier Series (FS)

FS is used to find the frequency components and their strengths for a given periodic signal $x(t)$.



The Three forms of Fourier Series

- Trigonometric Form
- Compact Trigonometric (Polar) Form.
- Complex Exponential Form.

Trigonometric Form

- It is simply a linear combination of sines and cosines at multiples of its fundamental frequency, $f_0 = 1/T$.

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(2\pi f_0 n t) + \sum_{n=1}^{\infty} b_n \sin(2\pi f_0 n t)$$

- $\boldsymbol{a_0}$ counts for any dc offset in $x(t)$.
- $\boldsymbol{a_0}$, $\boldsymbol{a_n}$, and $\boldsymbol{b_n}$ are called the trigonometric Fourier Series Coefficients.
- The $\boldsymbol{n^{th}}$ harmonic frequency is $\boldsymbol{nf_0}$.

Trigonometric Form

- **How to evaluate the Fourier Series Coefficients (FSC) of $x(t)$?**

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(2\pi f_0 n t) + \sum_{n=1}^{\infty} b_n \sin(2\pi f_0 n t)$$

To find a_0 integrate both side of the equation over a full period

$$a_0 = \frac{1}{T_0} \int_{T_0} x(t)\,dt$$

Trigonometric Form

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(2\pi f_0 nt) + \sum_{n=1}^{\infty} b_n \sin(2\pi f_0 nt)$$

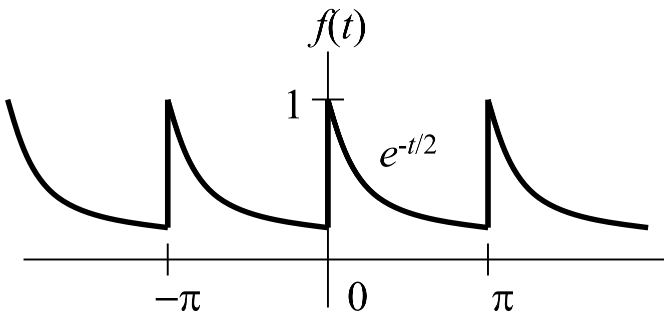
To find a_n multiply both side by $\cos(2\pi m f_0 t)$ and then integrate over a full period, $m = 1, 2, \ldots, n, \ldots \infty$

$$a_n = \frac{2}{T_0} \int_{T_0} x(t) \cos(2\pi f_0 nt)\, dt$$

To find b_n multiply both side by $\sin(2\pi m f_0 t)$ and then integrate over a full period, $m = 1, 2, \ldots, n, \ldots \infty$

$$b_n = \frac{2}{T_0} \int_{T_0} x(t) \sin(2\pi f_0 nt)\, dt$$

Example



$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(2nt) + b_n \sin(2nt)$$

$$a_0 = \frac{1}{\pi}\int_0^{\pi} e^{-\frac{t}{2}}\,dt = -\frac{2}{\pi}\left(e^{-\frac{\pi}{2}} - 1\right) \approx 0.504$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} e^{-\frac{t}{2}} \cos(2nt)\,dt = 0.504\left(\frac{2}{1+16n^2}\right)$$

$$b_n = \frac{2}{\pi}\int_0^{\pi} e^{-\frac{t}{2}} \sin(2nt)\,dt = 0.504\left(\frac{8n}{1+16n^2}\right)$$

a_n and b_n decrease in amplitude as $n \to \infty$.

- **Fundamental period**

  $T_0 = \pi$
- **Fundamental frequency**

  $f_0 = 1/T_0 = 1/\pi$ **Hz**

  $\omega_0 = 2\pi/T_0 = 2$ **rad/s**

$$f(t) = 0.504\left[1 + \sum_{n=1}^{\infty} \frac{2}{1+16n^2}\left(\cos(2nt) + 4n\sin(2nt)\right)\right]$$

To what value does the FS converge at the point of discontinuity?

Simplifications Through Signal Symmetry

- If $x(t)$ is EVEN: It must contain Cosine Terms and it may contain DC. Hence $b_n = 0$.

- If $x(t)$ is ODD: It must contain ONLY Sines Terms. Hence $a_0 = a_n = 0$.

Dirichlet Conditions

A periodic signal $x(t)$, has a Fourier series if it satisfies the following conditions:

1. $x(t)$ is **absolutely integrable** over any period, namely

$$\int_{T_0} |x(t)| dt < \infty$$

2. $x(t)$ has only a **finite number of maxima and minima** over any period

3. $x(t)$ has only a **finite number of discontinuities** over any period

Compact Trigonometric Form

- **Using single sinusoid,**

$$x(t) = \underbrace{C_0}_{\text{dc component}} + \sum_{n=1}^{\infty} \underbrace{C_n \cos(2\pi f_0 nt + \theta_n)}_{\text{nth harmonic}}$$

$$C_0 = a_0$$

- C_n, and θ_n **are related to the trigonometric coefficients a_n and b_n as:**

$$C_n = \sqrt{a_n^{\,2} + b_n^{\,2}}$$

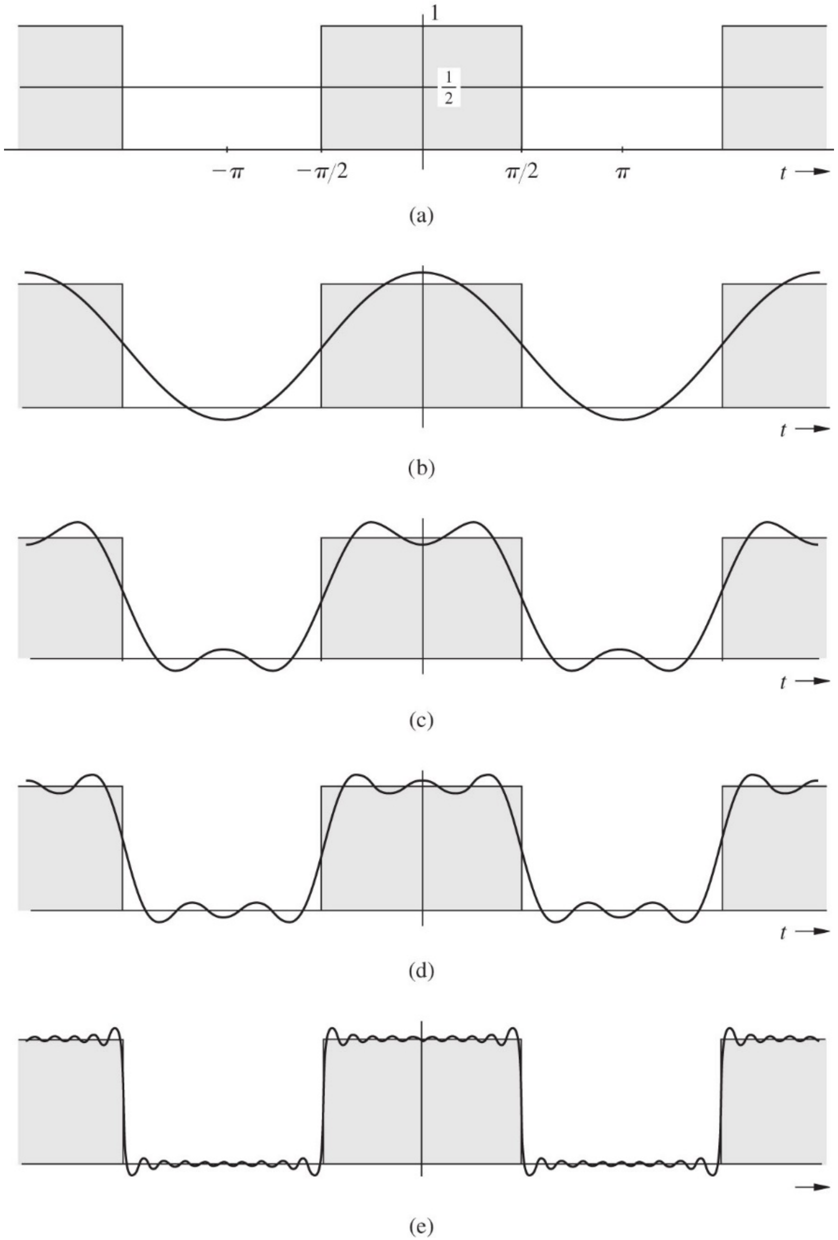
and

$$\theta_n = -\tan^{-1}\left(\frac{b_n}{a_n}\right)$$

The above relationships are obtained from the trigonometric identity

$$a\cos(x) + b\sin(x) = c\cos(x + \theta)$$

Role of Amplitude in Shaping Waveform



$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(2\pi f_0 n t + \theta_n)$$

Role of the Phase in Shaping a Periodic Signal

$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(2\pi f_0 nt + \theta_n)$$

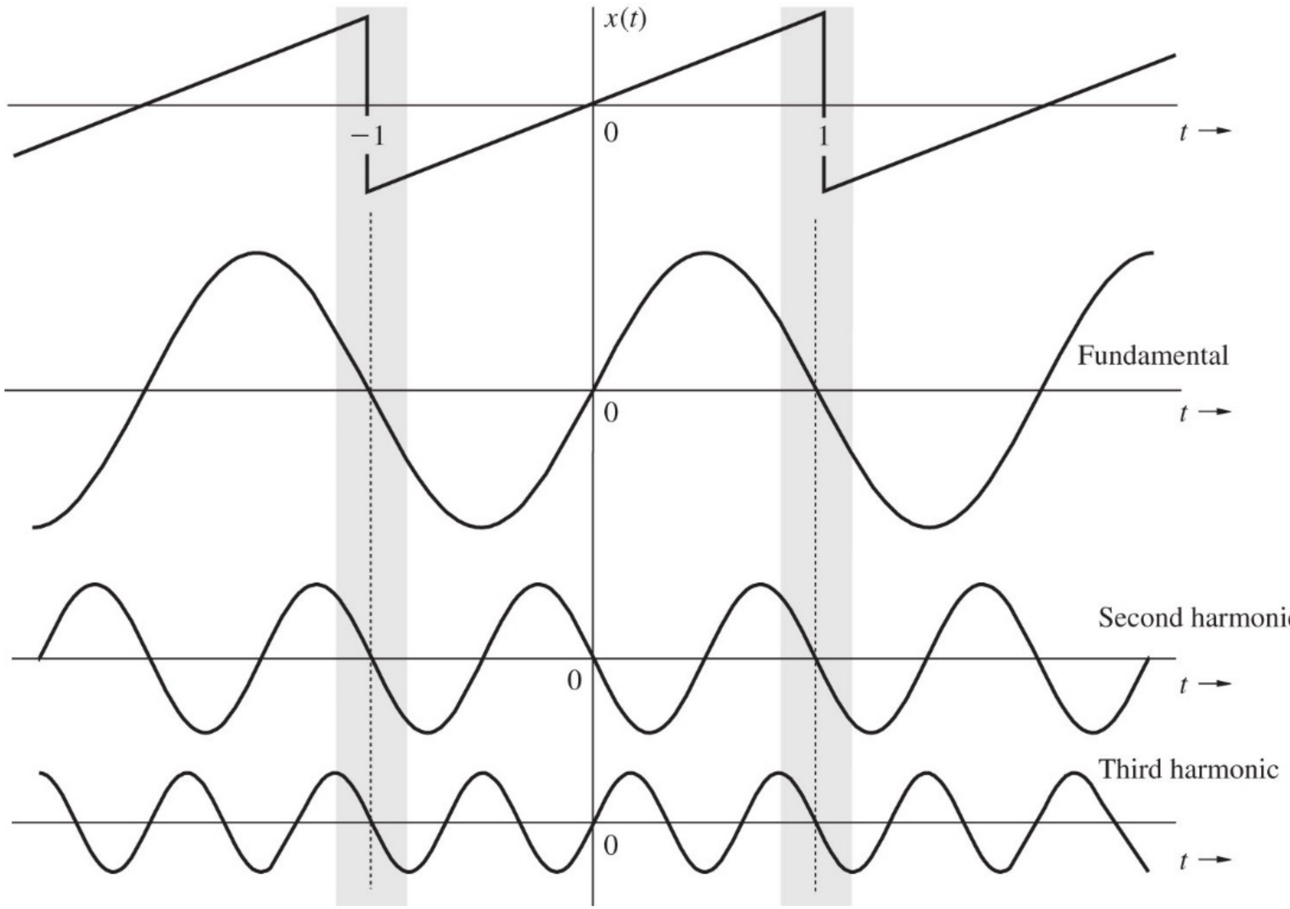
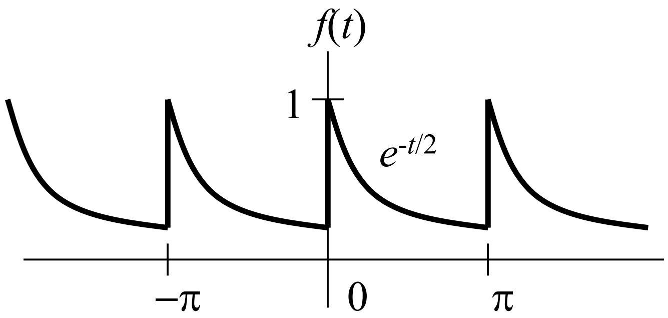


Figure 6.10 Role of the phase spectrum in shaping a periodic signal.

Compact Trigonometric



- **Fundamental period**

  $T_0 = \pi$
- **Fundamental frequency**

  $f_0 = 1/T_0 = 1/\pi$ Hz

  $\omega_0 = 2\pi/T_0 = 2$ rad/s

$$f(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(2nt - \theta_n)$$

$$a_0 \approx 0.504$$

$$a_n = 0.504\left(\frac{2}{1+16n^2}\right)$$

$$b_n = 0.504\left(\frac{8n}{1+16n^2}\right)$$

$$C_0 = a_o = 0.504$$

$$C_n = \sqrt{a_n^2 + b_n^2} = 0.504\left(\frac{2}{\sqrt{1+16n^2}}\right)$$

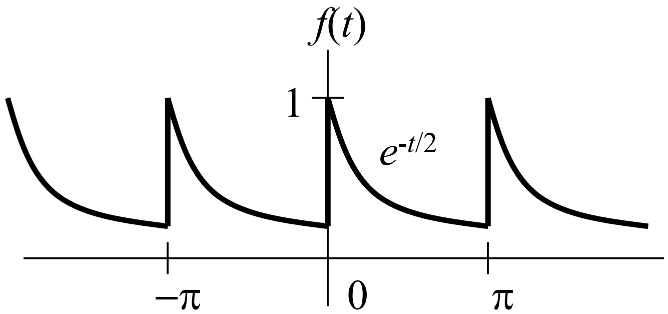
$$\theta_n = \tan^{-1}\left(\frac{-b_n}{a_n}\right) = -\tan^{-1} 4n$$

$$f(t) = 0.504 + 0.504\sum_{n=1}^{\infty}\frac{2}{\sqrt{1+16n^2}}\cos\left(2nt - \tan^{-1} 4n\right)$$

Line Spectra of x(t)

- The amplitude spectrum of $x(t)$ is defined as the plot of the magnitudes $|C_n|$ versus ω
- The phase spectrum of $x(t)$ is defined as the plot of the angles $\angle C_n = phase(C_n)$ versus ω
- This results in line spectra
- Bandwidth the difference between the highest and lowest frequencies of the spectral components of a signal.

Line Spectra

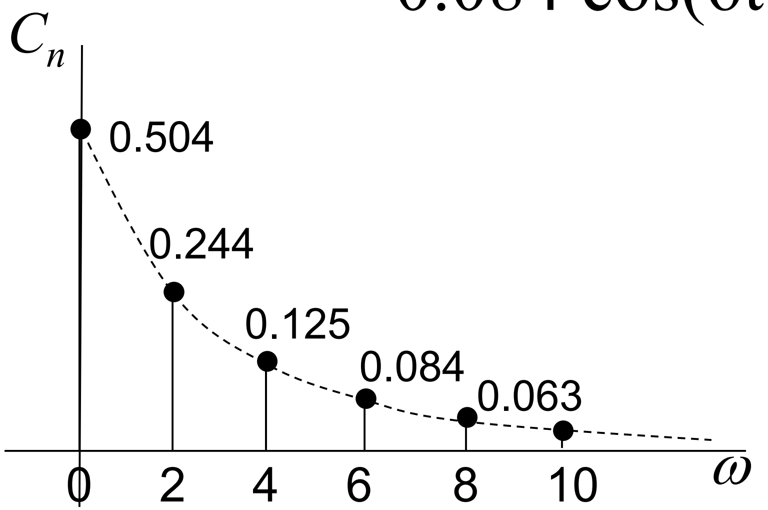


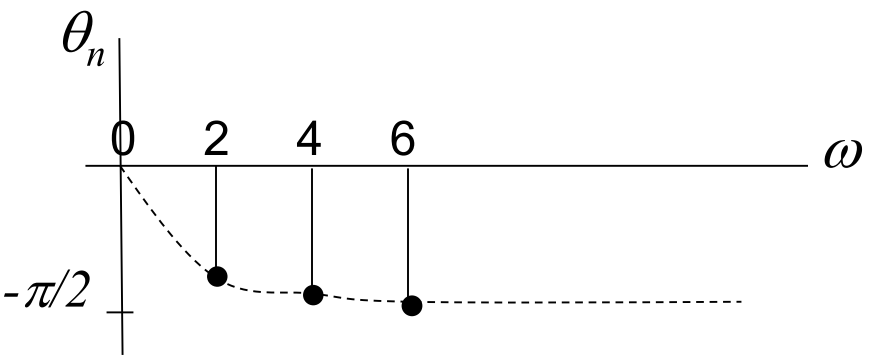
$$C_0 = 0.504 \qquad C_n = 0.504\left(\frac{2}{\sqrt{1+16n^2}}\right)$$

$$\theta_n = -\tan^{-1} 4n$$

$$f(t) = 0.504 + 0.504\sum_{n=1}^{\infty}\frac{2}{\sqrt{1+16n^2}}\cos\left(2nt - \tan^{-1} 4n\right)$$

$f(t)$=0.504 + 0.244 cos(2t-75.96°) + 0.125 cos(4t-82.87°) +
0.084 cos(6t-85.24°) + 0.063 cos(8t-86.24°) + …





Exponential Form

- $x(t)$ **can be expressed as**

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{j2\pi f_0 nt}$$

To find D_n multiply both side by $e^{-j2\pi f_0 nt}$ and then integrate over a full period, *m* =1,2,…,*n*,…∞

$$D_n = \frac{1}{T_o}\int_{T_o} x(t)e^{-j2\pi f_0 nt}\,dt \quad , \quad n = 0, \pm 1, \pm 2, \ldots.$$

D_n is a complex quantity in general $D_n = |D_n| e^{j\theta}$

$D_{-n} = D_n^*$ $\longrightarrow$ $|D_n| = |D_{-n}|$ $\longrightarrow$ $\angle D_n = -\angle D_{-n}$

Even **Odd**

D_0 is called the *constant or dc component* of $x(t)$

Line Spectra of $x(t)$ in the Exponential Form

- The ***line spectra*** for the exponential form has negative frequencies because of the mathematical nature of the complex exponent.

$$x(t) = \ldots + |D_{-2}|e^{-j\theta_2}e^{-j2\omega_0 t} + |D_{-1}|e^{-j\theta_1}e^{-j\omega_0 t} + D_0 +$$

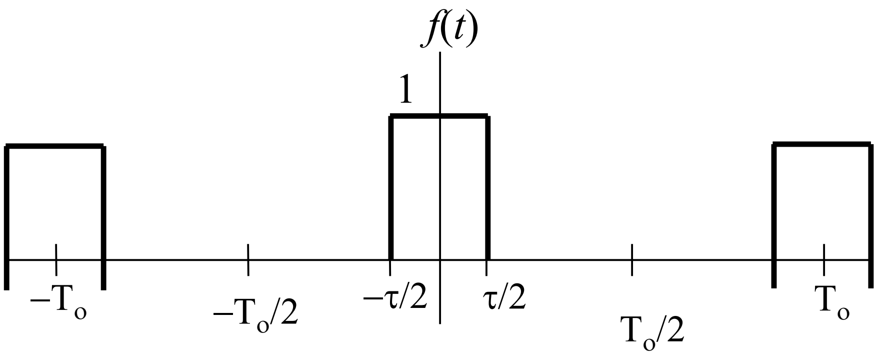
$$|D_1|e^{j\theta_1}e^{j\omega_0 t} + |D_2|e^{j\theta 2}e^{j2\omega_0 t} + \ldots$$

$$x(t) = C_0 + C_1\cos(\omega_0 t + \theta_1) + C_2\cos(2\omega_0 t + \theta_2) + \ldots$$

$C_0 = D_0$ $\qquad$ $C_n = 2|D_n|$ $\qquad$ $\angle C_n = \angle D_n$

Example

Find the exponential Fourier Series for the square-pulse periodic signal.



$$D_n = \frac{1}{T_0} \int_{-\tau/2}^{\tau/2} f(t) e^{-j\omega_o n t} dt$$

$$D_n = \frac{\tau}{T_0} sinc\left(\frac{n\omega_o \tau}{2}\right)$$

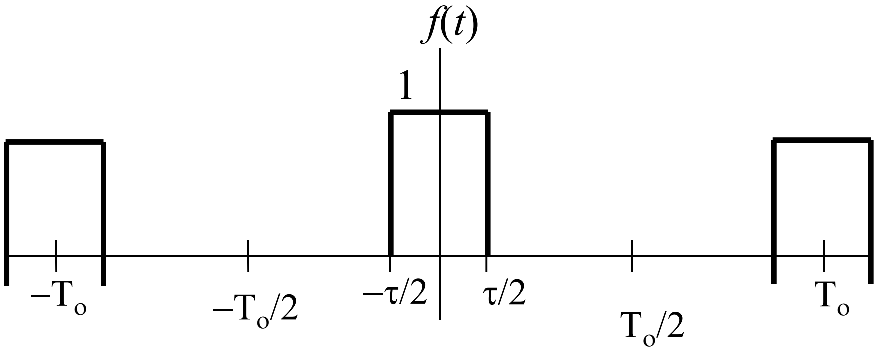
If $T_o = 2\pi$ and $\tau = \pi$ then

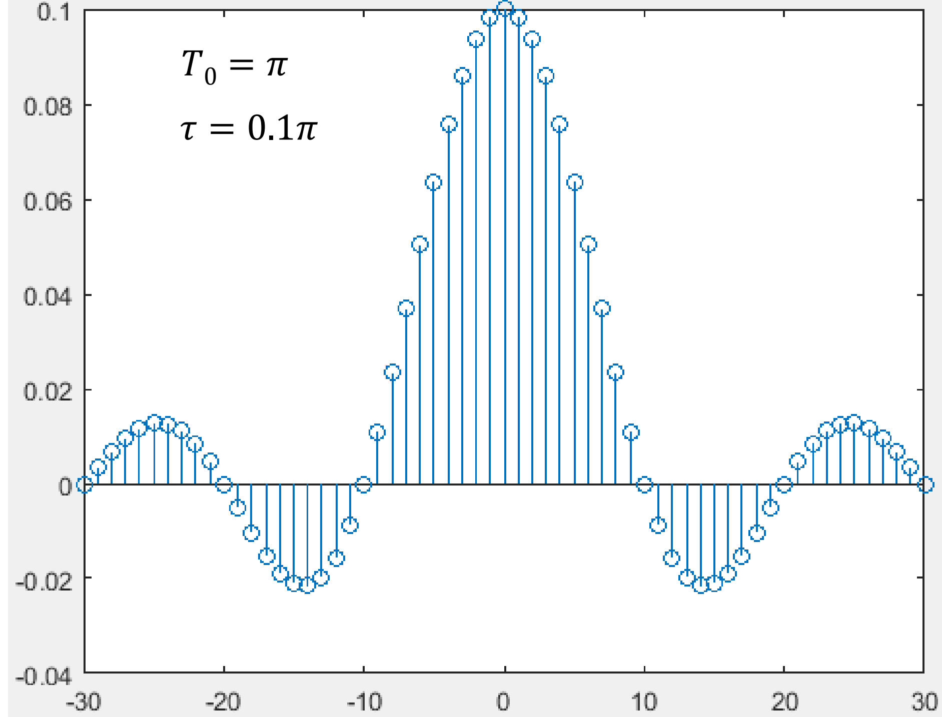
$$D_n = 0.5\text{sinc}(n\pi/2) \qquad D_0 = \frac{1}{2}$$

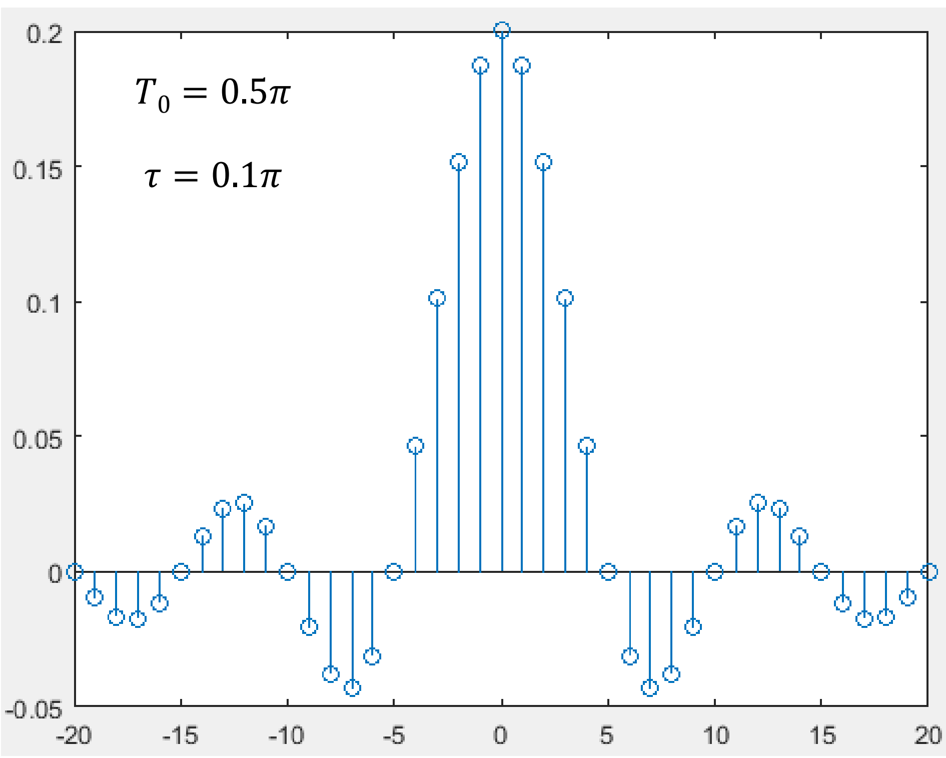
$$|D_n| = \begin{cases} 0 & n \text{ even} \\ 1/\pi n & n \text{ odd} \end{cases}$$

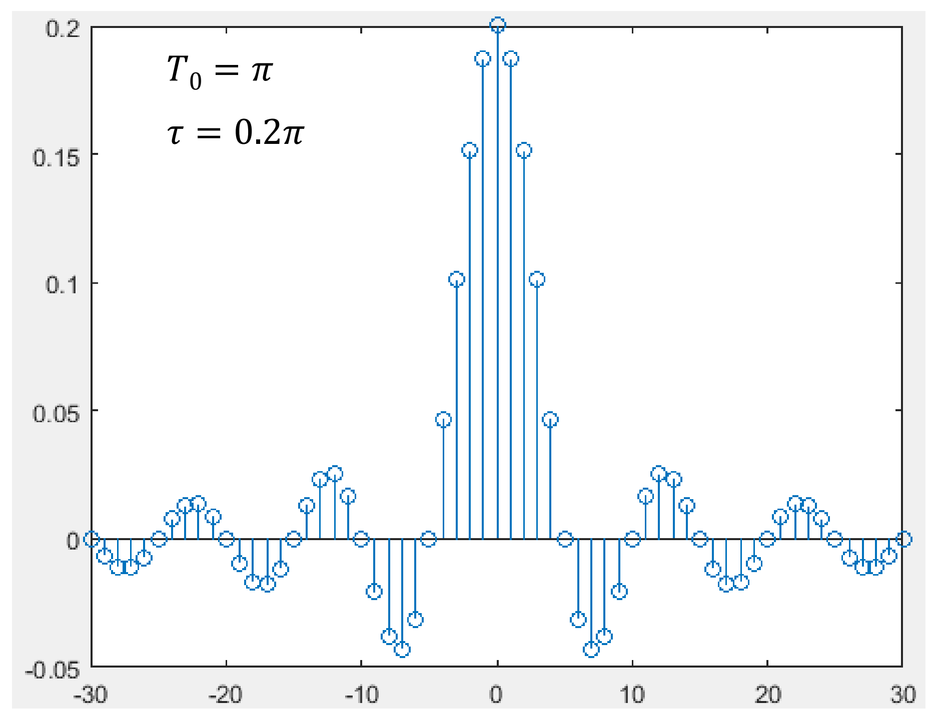
$$\theta_n = \begin{cases} 0 & \text{for all } n \neq 3,7,11,15,\ldots \\ -\pi & n = 3,7,11,15,\ldots \end{cases}$$

$$D_n = \frac{\tau}{T_0} sinc\left(\frac{n\omega_o \tau}{2}\right)$$

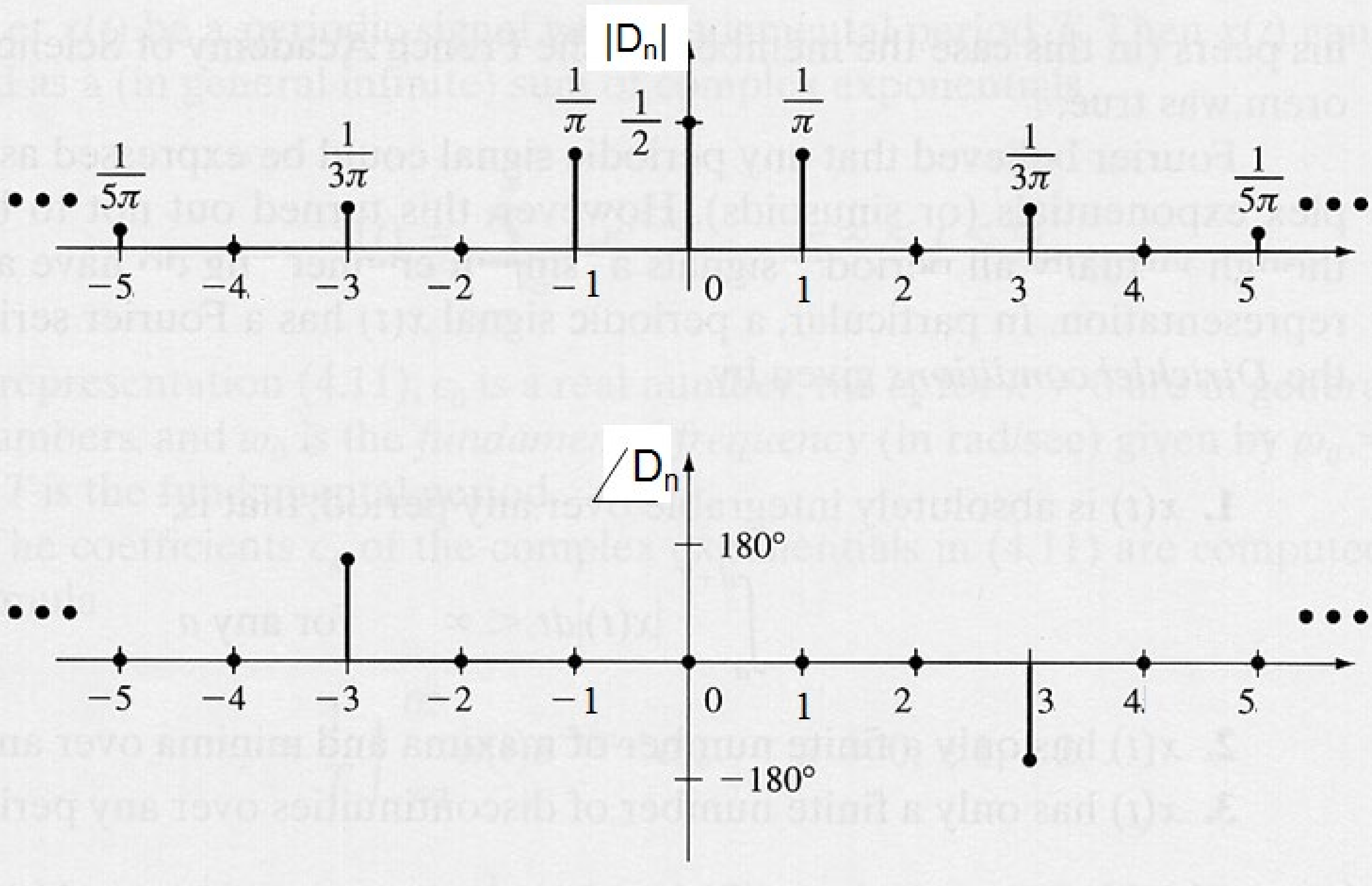








Exponential Line Spectra

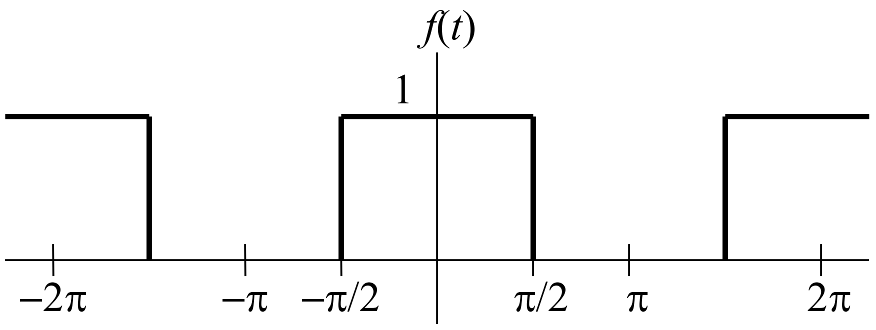


Line spectra for the rectangular pulse train.

Example

The compact trigonometric Fourier Series coefficients for the square-pulse periodic signal.

$$C_0 = \frac{1}{2}$$



$$C_n = \begin{cases} 0 & n \text{ even} \\ \dfrac{2}{\pi n} & n \text{ odd} \end{cases}$$

$$D_n = \frac{\tau}{T_0} sinc\left(\frac{n\omega_o \tau}{2}\right)$$

$$\theta_n = \begin{cases} 0 & \text{for all } n \neq 3,7,11,15,\dots \\ -\pi & n = 3,7,11,15,\dots \end{cases}$$

$$x(t) = \frac{1}{2} + \sum_{\substack{n=1 \\ odd}}^{\infty} \frac{2}{\pi n} \cos\left(nt + \left[(-1)^{|(n-1)/2|} - 1 \right] \frac{\pi}{2} \right)$$

Does the Fourier series converge to $x(t)$ at every point?

Gibbs Phenomenon

$x_3(t)$

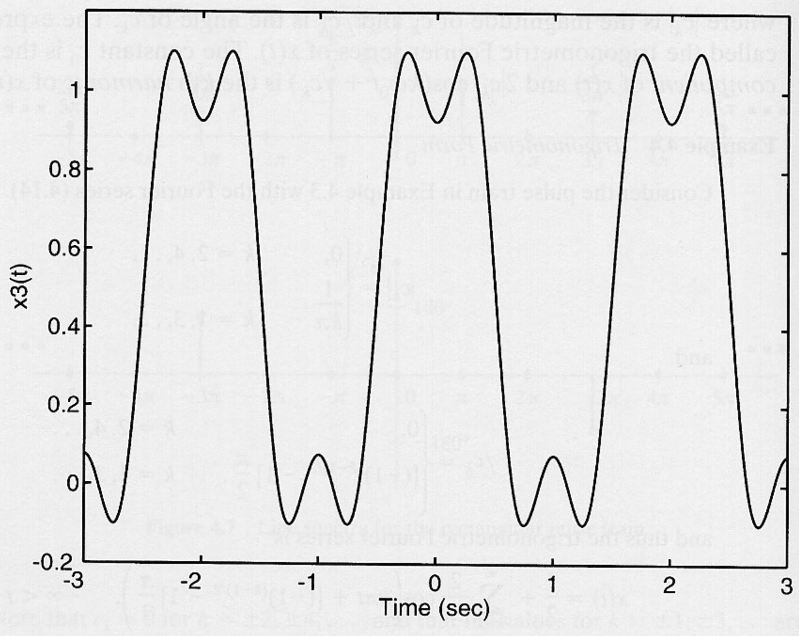


Figure 4.8 Plot of $x_N(t)$ when $N = 3$.

$x_9(t)$

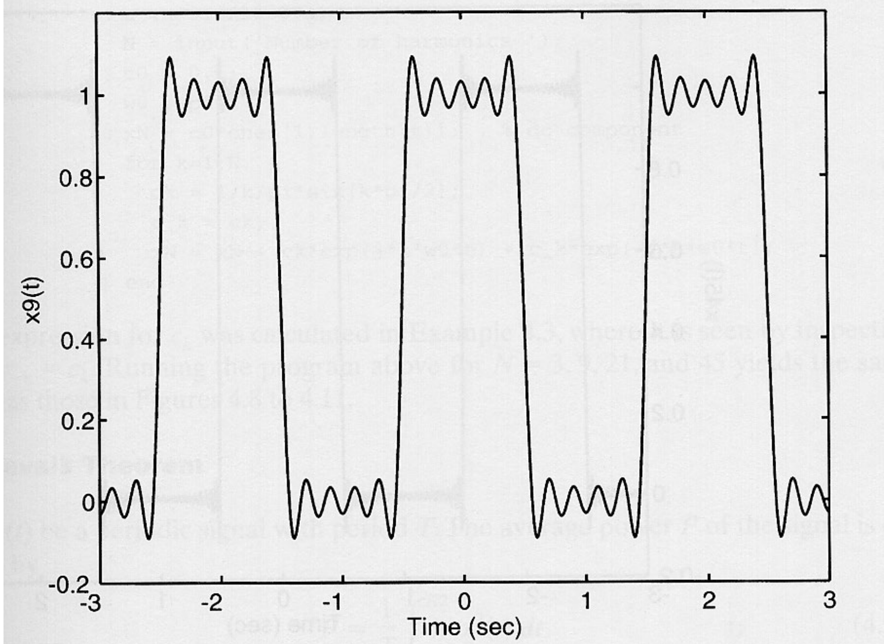


Figure 4.9 Approximation $x_9(t)$.

Gibbs Phenomenon – Cont'd

$x_{21}(t)$

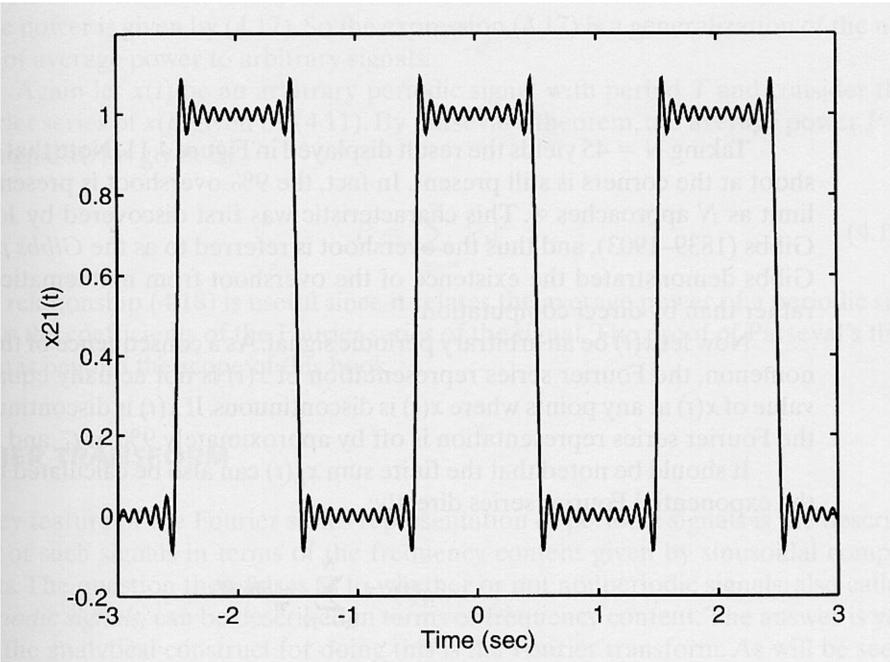


Figure 4.10 Approximation $x_{21}(t)$.

$x_{45}(t)$

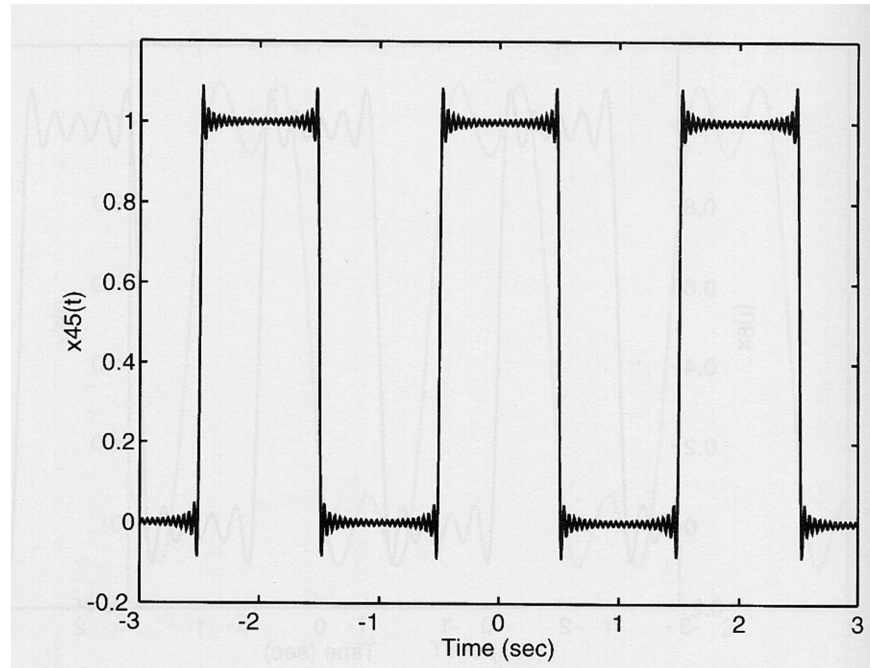
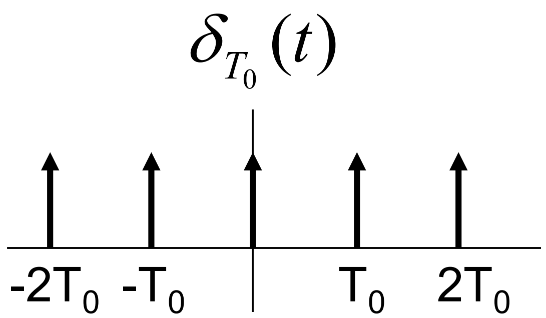


Figure 4.11 The signal $x_{45}(t)$.

overshoot: about 9 % of the signal magnitude (present even if $N \to \infty$)

Example

Find the exponential Fourier Series and sketch the corresponding spectra for the impulse train shown below. From this result sketch the trigonometric spectrum and write the trigonometric Fourier Series.



Solution

$$D_n = 1/T_0$$

$$\delta_{T_0}(t) = \frac{1}{T_0}\sum_{n=-\infty}^{\infty} e^{jn\omega_0 t}$$

$$C_n = 2|D_n| = 2/T_0$$

$$C_0 = |D_0| = 1/T_0$$

$$\delta_{T_0}(t) = \frac{1}{T_0}\left[1 + 2\sum_{n=1}^{\infty}\cos(n\omega_0 t)\right]$$

Relationships between the Coefficients of the Different Forms

$$D_n = 0.5\left(a_n - jb_n\right)$$
$$D_{-n} = D^*{}_n = 0.5\left(a_n + jb_n\right)$$
$$D_n = 0.5C_n\angle\theta_n = 0.5C_n e^{j\theta_n}$$
$$D_0 = a_0 = C_0$$

$$a_n = D_n + D_{-n} = 2\,\mathrm{Re}\left\{D_n\right\}$$
$$b_k = j\left(D_n - D_{-n}\right) = -2\,\mathrm{Im}\left\{D_n\right\}$$
$$a_n = C_n\cos\left(\theta_n\right)$$
$$b_n = -C_n\sin\left(\theta_n\right)$$
$$a_0 = D_0 = c_0$$

$$C_n = \sqrt{a_n{}^2 + b_n{}^2}$$
$$\theta_n = -\tan^{-1}\left(\frac{b_n}{a_n}\right)$$
$$C_n = 2\left|D_n\right|$$
$$\theta_n = \angle D_n$$
$$C_0 = a_0 = D_0$$

Parseval's Theorem

- Let $x(t)$ be a periodic signal with period *T*
- The ***average power*** *P* of the signal is defined as

$$P = \frac{1}{T}\int_{-T/2}^{T/2} \left|x(t)\right|^2 dt$$

- Expressing the signal as

$$x(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n\omega_0 t + \theta_n)$$

it is also

$$P = C_0{}^2 + \sum_{n=1}^{\infty} 0.5 C_n{}^2 \qquad\qquad P = D_0^2 + 2\sum_{n=1}^{\infty}\left|D_n\right|^2$$

LTIC System Response to Periodic Inputs

$$e^{j\omega_0 t} \longrightarrow \boxed{\begin{matrix} H(s) \\ H(j\omega) \end{matrix}} \longrightarrow H(j\omega_0)e^{j\omega_0 t}$$

A periodic signal $x(t)$ with period T_0 can be expressed as

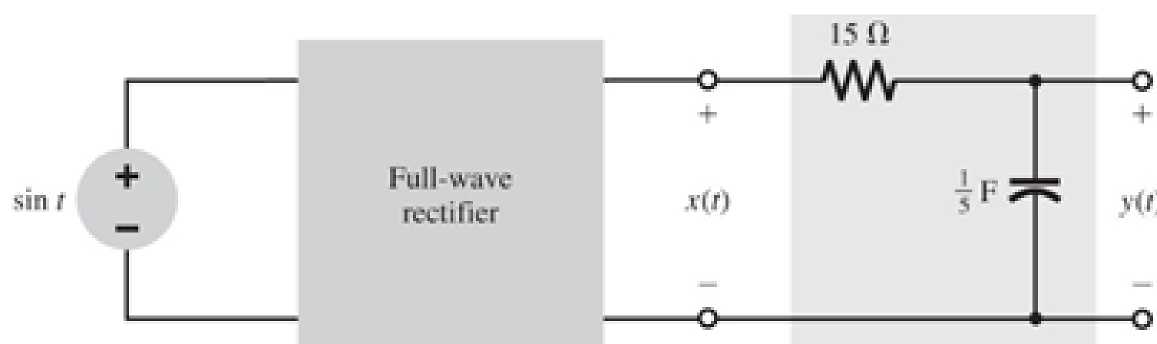
$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$$

For a linear system

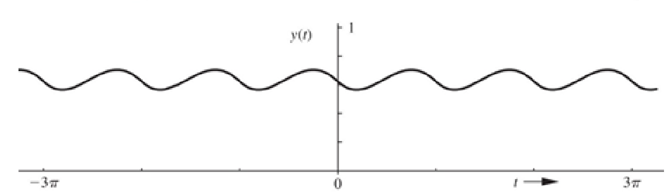
$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} \longrightarrow \boxed{\begin{matrix} H(s) \\ H(j\omega) \end{matrix}} \longrightarrow y(t) = \sum_{n=-\infty}^{\infty} H(jn\omega_0) D_n e^{jn\omega_0 t}$$

Fourier Series Analysis of DC Power Supply

A full-wave rectifier is used to obtain a dc signal from a sinusoid $sin(t)$. The rectified signal $x(t)$ is applied to the input of a lowpass RC filter, which suppress the time-varying component and yields a dc component with some residual ripple. Find the filter output $y(t)$. Find also the dc output and the rms value of the ripple voltage.



$$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2}(a \sin bx - b \cos bx)$$



Fourier Series Analysis of DC Power Supply

$$D_n = \frac{2}{\pi(1-4n^2)}$$

$$x(t) = \sum_{n=-\infty}^{\infty} \frac{2}{\pi(1-4n^2)} e^{j2nt}$$

$$H(j\omega) = \frac{1}{3j\omega+1}$$

$$y(t) = \sum_{n=-\infty}^{\infty} D_n H(jn\omega_0) e^{jn\omega_0 t}$$

$$y(t) = \sum_{n=-\infty}^{\infty} \frac{2}{\pi(1-4n^2)(j6n+1)} e^{j2nt}$$

$$D_0 = 2/\pi \qquad P_{DC} = 4/\pi^2$$

$$P_{ripple} = 2\sum_{n=1}^{\infty} |D_n|^2$$

$$|D_n| = \frac{2}{\pi(1-4n^2)\sqrt{36n^2+1}}$$

$$P_{ripple} = 0.0025$$

$$\text{ripple rms} = \sqrt{P_{ripple}} = 0.05$$

Ripple rms is only 5% of the input amplitude

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```
clear all
t=0:1/1000:3*pi;
for i=1:100
    n=i;
    yp=(2*exp(j*2*n*t))/(pi*(1-4*n^2)*(j*6*n+1));
    n=-i;
    yn=(2*exp(j*2*n*t))/(pi*(1-4*n^2)*(j*6*n+1));
    y(i,:)=yp+yn;
end
yf = 2/pi + sum(y);
plot(t,yf, t, (2/pi)*ones(1,length(yf)))
axis([0 3*pi 0 1]);

Power=0;
for n=1:50
    Power(n) = abs(2/(pi*(1-4*n^2)*(j*6*n+1)));
end
TotalPower = 2*sum((Power.^2));
figure; stem( Power(1,1:20));
```

This Matlab code will plot y(t) for $-100 \le n \le 100$ and find the ripple power according to the equations below

$$y(t) = \sum_{n=-\infty}^{\infty} \frac{2}{\pi(1-4n^2)(j6n+1)} e^{j2nt}$$

$$P_{ripple} = 2\sum_{n=1}^{\infty} |D_n|^2 = 0.0025$$